

System1 classifier comparisons for customer-care transcripts

Open encoders · TypeSafe.ai JEV · GPT-5.6 Sol and Luna

99.93%

Best accuracy

GPT-5.6 Sol

99.90%

JEV calibrated

4 / 4,050 errors

99.16%

Best local encoder

ModernBERT trained

1,000 synthetic conversations · 27 attributes · 700 train / 150 validation / 150 locked test

Mark Austin

Independent evaluation · September 20, 2026

Executive findings

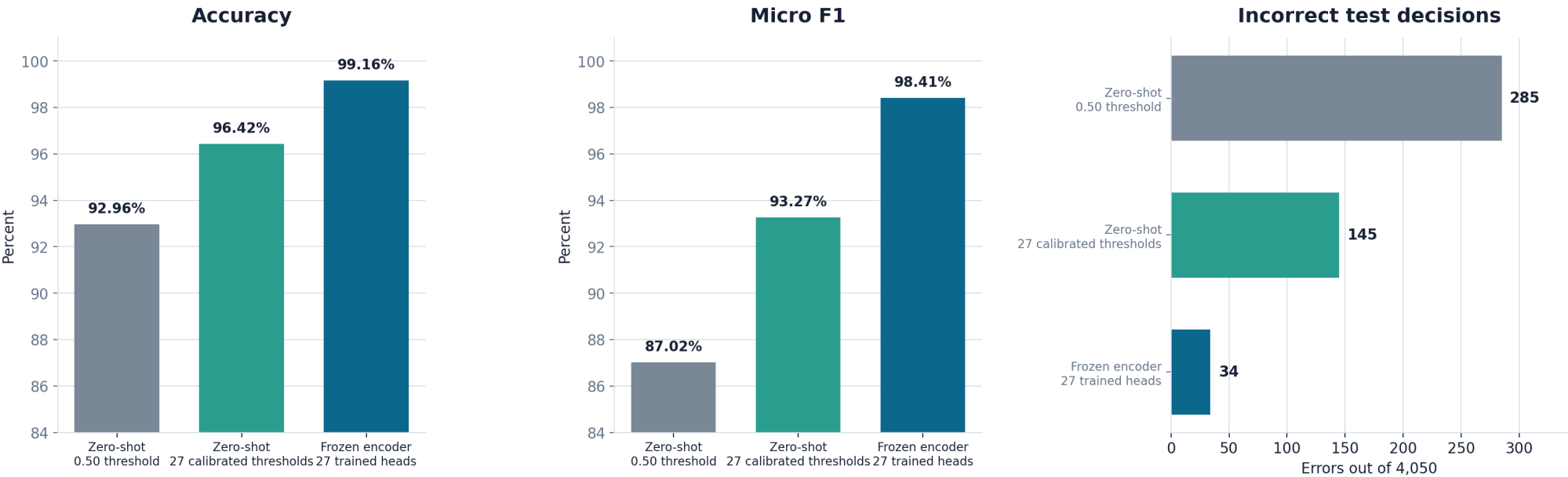
Quality, calibration, and operating-cost implications

- Sol achieved the highest measured quality: 99.93% accuracy and 99.86% micro F1, with three errors across 4,050 held-out decisions.
- Validation-calibrated JEV Noul was one decision behind Sol: 99.90% accuracy and 99.81% F1. Its estimated recurring API charge is about \$0.117 per 1,000 transcripts under the measured prompt shape.
- The strongest fully local result was frozen ModernBERT plus 27 supervised logistic heads: 99.16% accuracy and 98.41% F1.
- Threshold calibration alone improved unchanged zero-shot ModernBERT from 92.96% to 96.42% accuracy and cut errors from 285 to 145.
- GLiClass Modern Large v3 was the strongest zero-shot open encoder: validation-only calibration reached 97.90% accuracy and 96.00% F1.
- DeBERTa-v3-large zero-shot (-c) did not beat ModernBERT zero-shot and was approximately 2.47x slower on the measured CPU path.
- At the 224-token benchmark mean, simulated cost per 1,000 transcripts is \$0.0018 for trained ModernBERT, \$0.0043 for GLiClass Modern, \$0.0525 for zero-shot ModernBERT, and \$0.1169 for JEV Noul.
- The standardized one-call API estimates are \$0.4000 for Luna and \$7.1520 for Sol per 1,000 benchmark-mean transcripts; hidden reasoning tokens are excluded.

Interpret carefully. This is a synthetic, template-driven smoke test. It is useful for controlled comparisons and pipeline validation, but it does not establish production accuracy. The JEV Choice and Noul runs also differ in state structure and criteria specificity, so their gap is not a primitive-only A/B result.

Threshold calibration captures much of training's gain without changing model weights

ModernBERT-large · same 4,050 locked test decisions · calibration closes 55.8% of the accuracy gap and 54.8% of the F1 gap

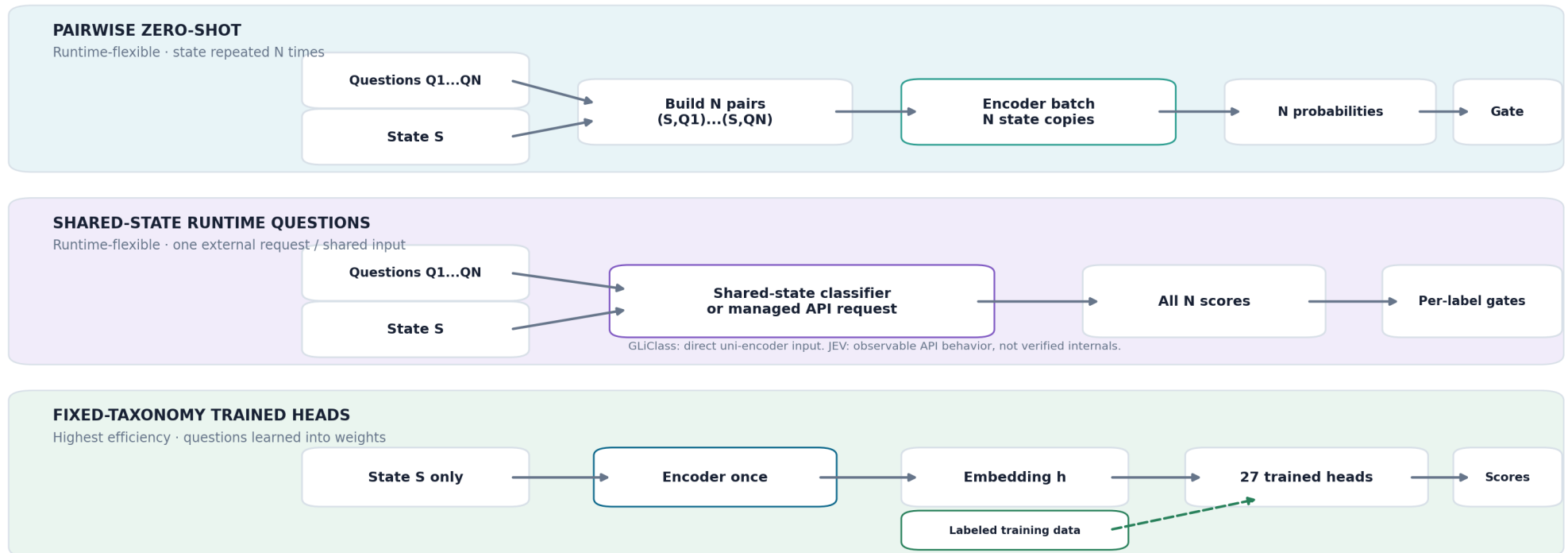


Calibration changes only 27 scalar cutoffs: accuracy rises 3.46 points and errors fall from 285 to 145. Trained heads learn 27 x (1,024 weights + intercept), reaching 34 errors.

The zero-shot architecture still accepts new questions at runtime, but a new question does not inherit a validated label-specific threshold. Trained heads require labeled data and retraining.

Three ways to combine state and classification questions

The efficiency/flexibility trade-off is determined by where question meaning enters the system



Calibration changes gates after scoring; it does not change whether the state is repeated, shared, or encoded once for fixed learned heads.

Fixed taxonomy: which approach has the lowest modeled cost?

Trained ModernBERT heads are eligible · state length × taxonomy size × paid GPU utilization

ModernBERT-large + trained heads TypeSafe.ai JEV Noul

10% paid H100 utilization

10	Trained MB \$0.082	JEV \$0.208	JEV \$0.376	JEV \$0.712	JEV \$1.434
25	Trained MB \$0.082	JEV \$0.268	JEV \$0.436	JEV \$0.772	JEV \$1.554
50	Trained MB \$0.082	Trained MB \$0.326	JEV \$0.535	JEV \$0.871	JEV \$1.753
100	Trained MB \$0.082	Trained MB \$0.326	Trained MB \$0.652	JEV \$1.070	JEV \$2.151

25% paid H100 utilization

10	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
25	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
50	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
100	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077

50% paid H100 utilization

10	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
25	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
50	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
100	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
	1k	4k	8k	16k	32k

100% paid H100 utilization

10	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
25	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
50	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
100	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
	1k	4k	8k	16k	32k

Each cell shows the winner and USD per 1,000 states. ModernBERT-large + trained heads encodes the state once and has negligible incremental head cost, but new or changed questions require labeled data and retraining.

Runtime questions: lowest-cost approach

Trained ModernBERT heads are ineligible · state length × question count × paid GPU utilization

TypeSafe.ai JEV Noul GLiClass Modern Large v3

10% paid H100 utilization

Runtime questions	10	JEV \$0.082	JEV \$0.208	JEV \$0.376	JEV \$0.712	JEV \$1.434
	25	GLiClass \$0.104	JEV \$0.268	JEV \$0.436	JEV \$0.772	JEV \$1.554
	50	GLiClass \$0.126	JEV \$0.367	JEV \$0.535	JEV \$0.871	JEV \$1.753
	100	GLiClass \$0.171	GLiClass \$0.416	JEV \$0.734	JEV \$1.070	JEV \$2.151

25% paid H100 utilization

GLiClass \$0.036	GLiClass \$0.134	GLiClass \$0.265	GLiClass \$0.549	GLiClass \$1.095
GLiClass \$0.042	GLiClass \$0.139	GLiClass \$0.287	GLiClass \$0.565	GLiClass \$1.122
GLiClass \$0.051	GLiClass \$0.148	GLiClass \$0.305	GLiClass \$0.592	GLiClass \$1.167
GLiClass \$0.068	GLiClass \$0.166	GLiClass \$0.341	GLiClass \$0.646	GLiClass \$1.257

50% paid H100 utilization

Runtime questions	10	GLiClass \$0.018	GLiClass \$0.067	GLiClass \$0.132	GLiClass \$0.275	GLiClass \$0.548
	25	GLiClass \$0.021	GLiClass \$0.070	GLiClass \$0.144	GLiClass \$0.283	GLiClass \$0.561
	50	GLiClass \$0.025	GLiClass \$0.074	GLiClass \$0.153	GLiClass \$0.296	GLiClass \$0.583
	100	GLiClass \$0.034	GLiClass \$0.083	GLiClass \$0.171	GLiClass \$0.323	GLiClass \$0.628
		1k	4k	8k	16k	32k

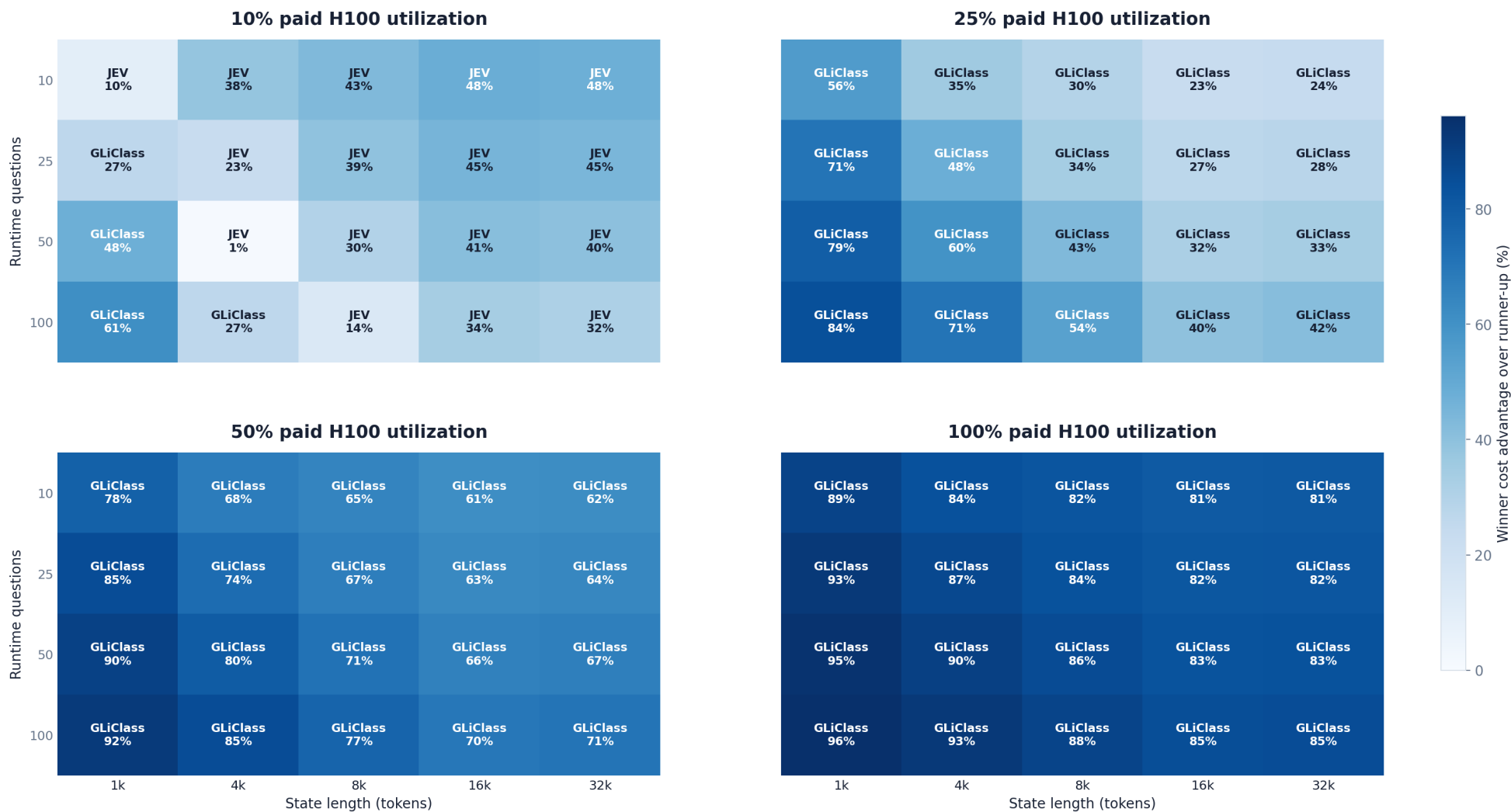
100% paid H100 utilization

GLiClass \$0.009	GLiClass \$0.034	GLiClass \$0.066	GLiClass \$0.137	GLiClass \$0.274
GLiClass \$0.010	GLiClass \$0.035	GLiClass \$0.072	GLiClass \$0.141	GLiClass \$0.280
GLiClass \$0.013	GLiClass \$0.037	GLiClass \$0.076	GLiClass \$0.148	GLiClass \$0.292
GLiClass \$0.017	GLiClass \$0.042	GLiClass \$0.085	GLiClass \$0.162	GLiClass \$0.314
1k	4k	8k	16k	32k

Each cell shows the winner and USD per 1,000 states. Runtime candidates evaluated: GLiClass Modern, TypeSafe.ai JEV Noul, ModernBERT NLI, Luna, and Sol. Use the paired fixed-taxonomy map when trained heads are acceptable.

How decisive is the modeled cost winner?

Runtime questions only; trained heads are ineligible · high percentages are robust, low percentages indicate a close decision



Advantage = (runner-up cost - winner cost) / runner-up cost. Operational simplicity, data residency, latency, support, and quality may outweigh a small modeled cost advantage.

Executive decision map: economics versus runtime flexibility

Reference workload: 6,000 state tokens × 25 questions · self-hosted cost at 100% H100 utilization

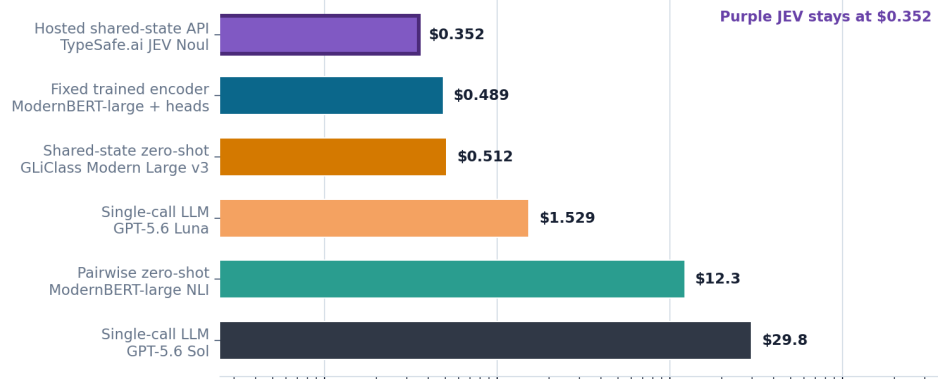


The \$1/1,000 boundary is an executive guide, not a universal purchasing threshold. Flexible systems accept new questions without retraining; trained heads do not.

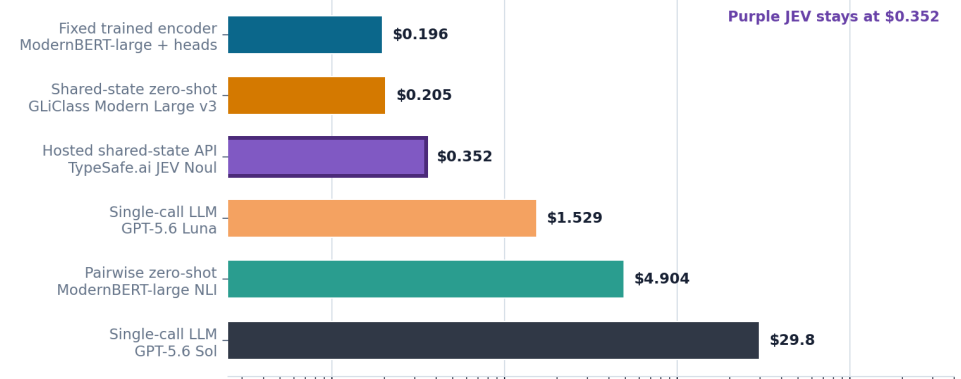
GPU utilization determines self-hosted encoder economics

Fixed workload: 6k state tokens × 25 questions · NVIDIA H100 at USD 5/hour · lower is better

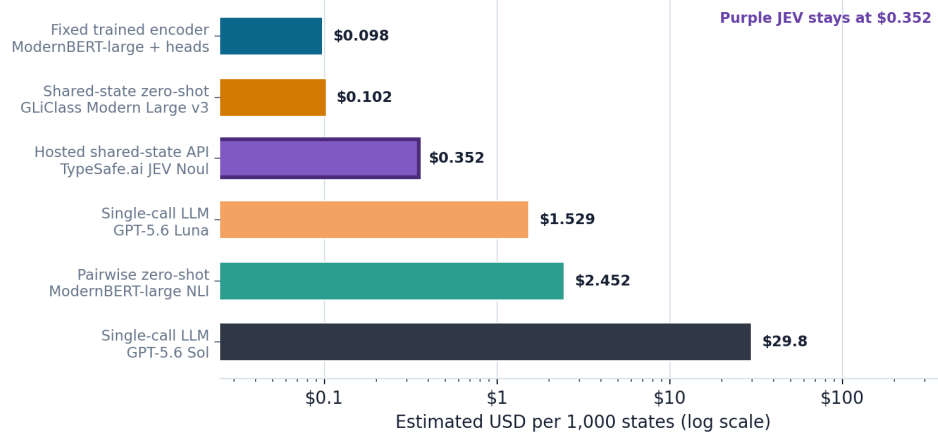
10% paid H100 utilization



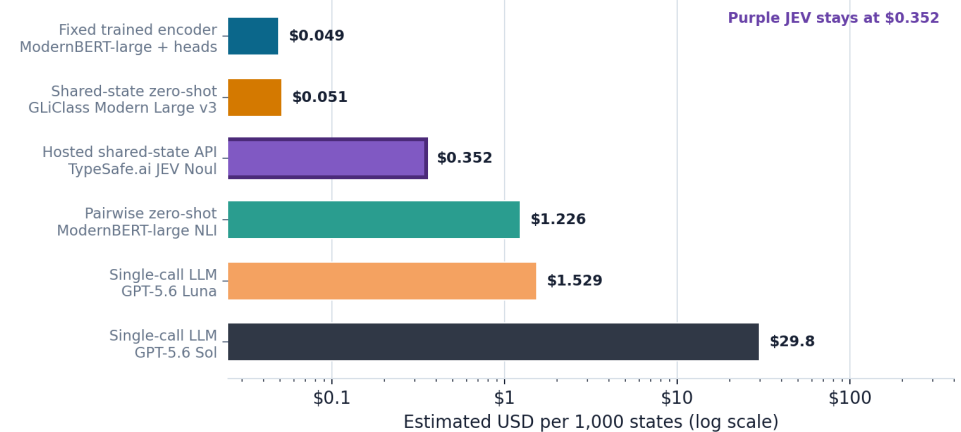
25% paid H100 utilization



50% paid H100 utilization



100% paid H100 utilization



Self-hosted encoder cost is divided by paid GPU utilization; hosted JEV/Luna/Sol prices remain usage-based. In this scenario, JEV crosses trained ModernBERT near 13.9% utilization and GLiClass Modern near 14.6%. Hidden LLM reasoning tokens are excluded.

Operating-model scorecard

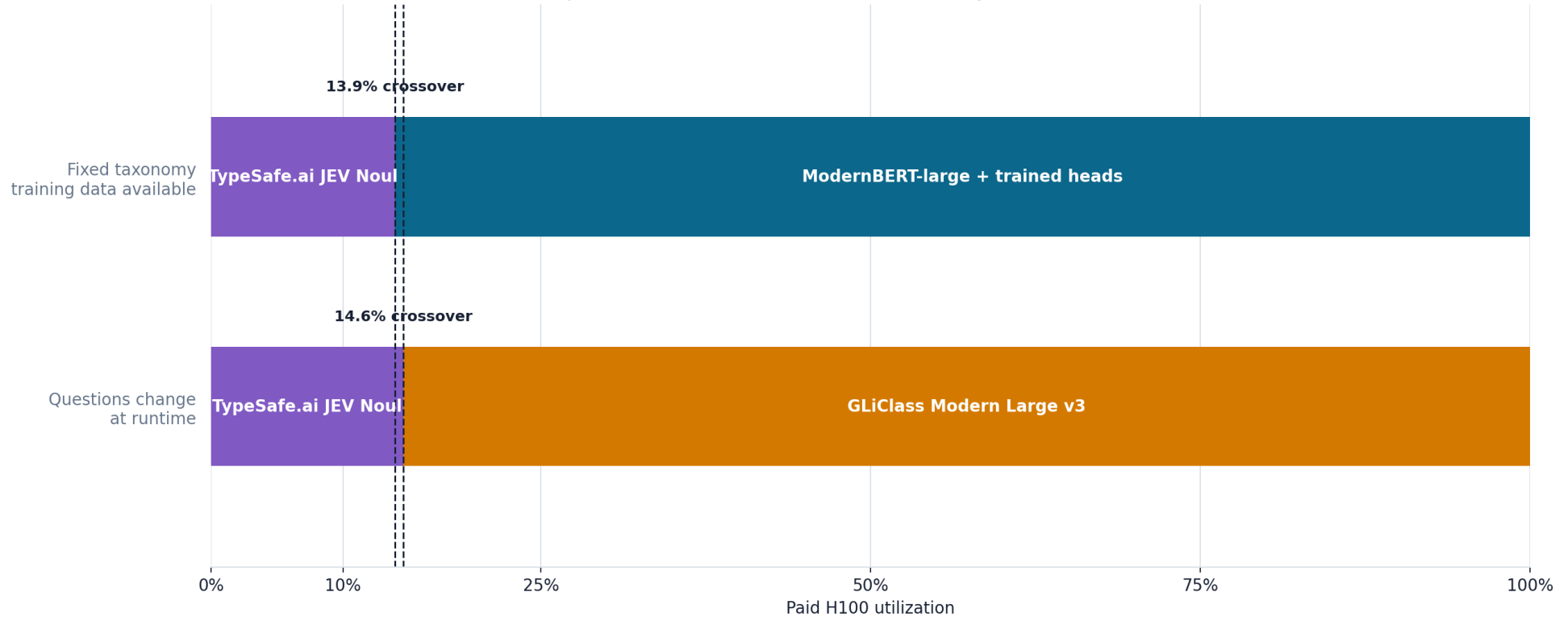
Named examples from this benchmark · choose the operating model before choosing the model

Operating pattern	Example implementation	Accuracy	Runtime questions	Deployment	Executive takeaway
Fixed trained encoder	ModernBERT-large + trained logistic heads	99.16%	No	Self-hosted	Default economic path when taxonomy is stable
Shared-state zero-shot	GLiClass Modern Large v3	97.90%	Yes	Self-hosted	Flexible winner above ~14.6% utilization at 6k × 25
Hosted shared-state API	TypeSafe.ai JEV Noul	99.90%	Yes	Managed API	Usage-priced; no customer-owned idle GPU
Pairwise zero-shot	ModernBERT-large zero-shot NLI	96.42%	Yes	Self-hosted	Repeats state for every question
Single-call LLM	GPT-5.6 Luna / GPT-5.6 Sol	99.75% / 99.93%	Yes	Managed API	Premium option for broader semantic reasoning

Accuracy is measured on the locked 150-transcript synthetic test. The 14.6% utilization crossover is specific to the 6k-state-token × 25-question, USD 5/H100-hour scenario.

Which operating model wins as GPU utilization changes?

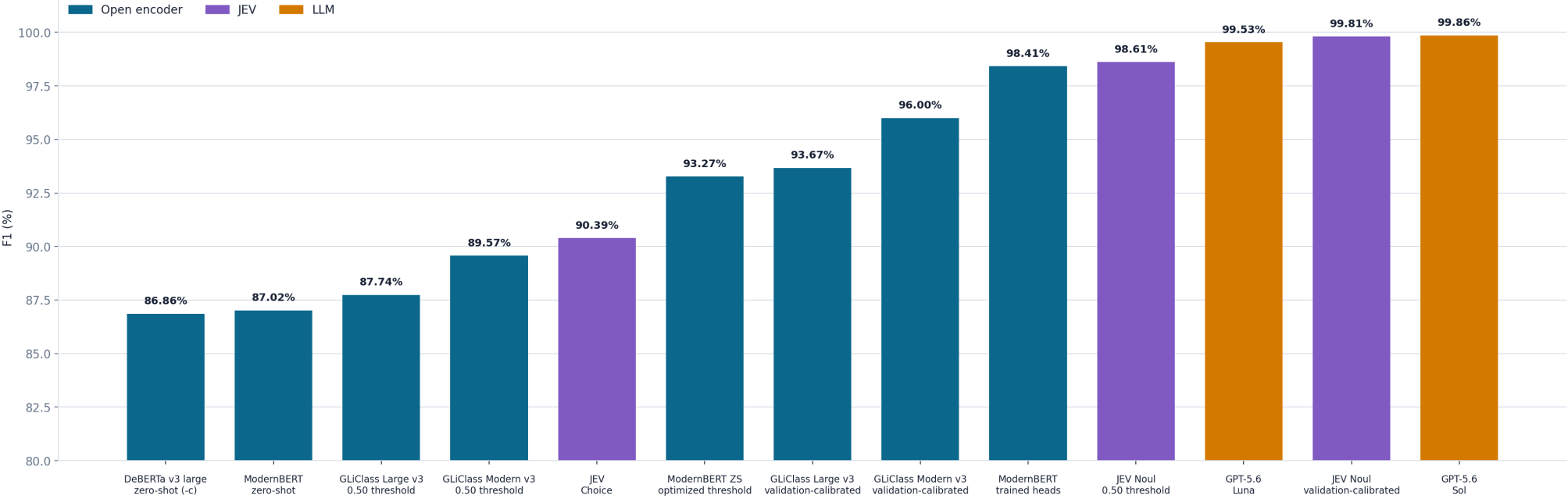
Reference workload: 6k state tokens × 25 questions · H100 at USD 5/hour · modeled cost only



Below the crossover, usage-priced JEV is cheaper than carrying idle H100 capacity. Above it, self-hosted encoders win on modeled marginal cost. Operational requirements and measured quality remain separate gates.

F1 comparison across transcript classifiers

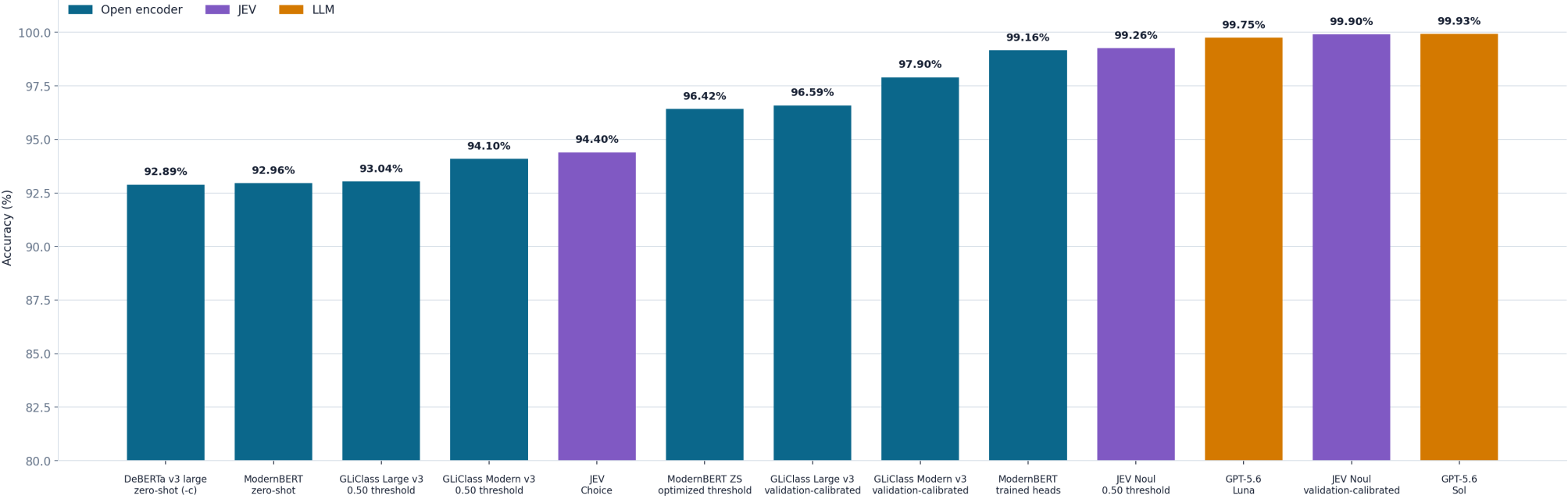
150 locked test transcripts · 27 binary attributes · models ordered by accuracy



F1 axis begins at 80% to make differences legible.

Accuracy comparison across transcript classifiers

150 locked test transcripts · 27 binary attributes · models ordered by accuracy



Accuracy axis begins at 80% to make differences legible.

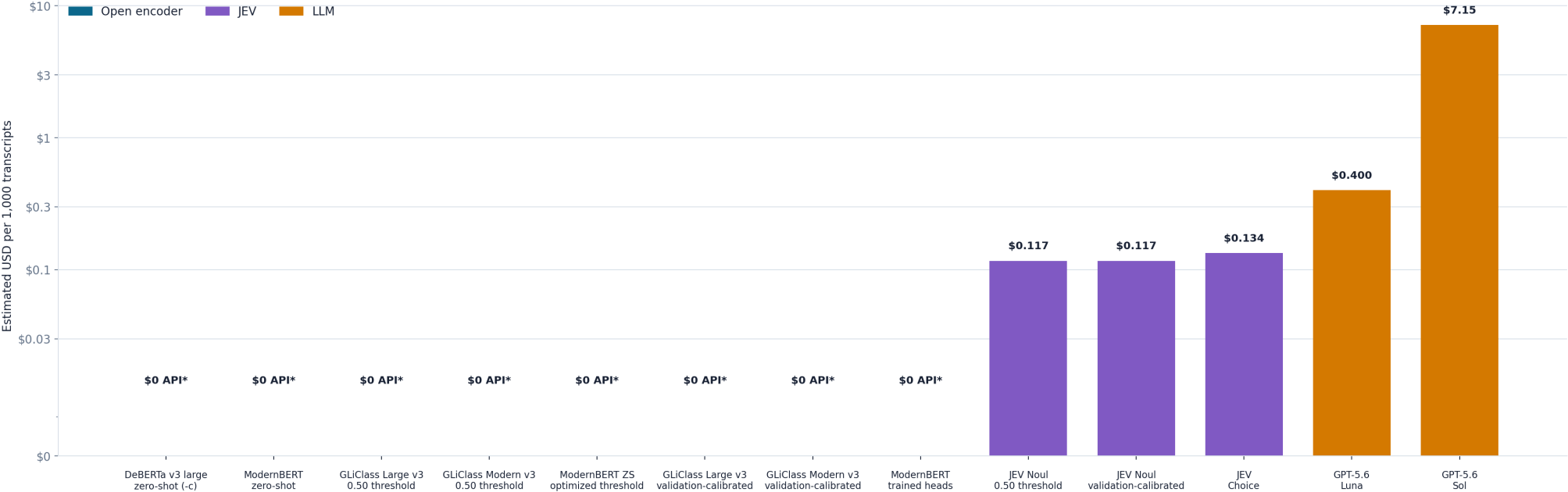
All transcript-classification quality metrics

Held-out test set · micro metrics across 4,050 decisions; exact match requires all 27 labels correct

Approach	Family	Accuracy	Precision	Recall	F1	Exact match
DeBERTa v3 large zero-shot (-c)	Open encoder	92.89%	84.70%	89.14%	86.86%	13.33%
ModernBERT zero-shot	Open encoder	92.96%	84.74%	89.42%	87.02%	14.00%
GLiClass Large v3 0.50 threshold	Open encoder	93.04%	81.90%	94.48%	87.74%	12.00%
GLiClass Modern v3 0.50 threshold	Open encoder	94.10%	83.89%	96.07%	89.57%	16.67%
JEV Choice	JEV	94.40%	82.47%	100.00%	90.39%	18.00%
ModernBERT ZS optimized threshold	Open encoder	96.42%	92.53%	94.01%	93.27%	34.67%
GLiClass Large v3 validation-calibrated	Open encoder	96.59%	91.82%	95.60%	93.67%	34.67%
GLiClass Modern v3 validation-calibrated	Open encoder	97.90%	96.41%	95.60%	96.00%	53.33%
ModernBERT trained heads	Open encoder	99.16%	98.05%	98.78%	98.41%	83.33%
JEV Noul 0.50 threshold	JEV	99.26%	97.27%	100.00%	98.61%	80.67%
GPT-5.6 Luna	LLM	99.75%	99.07%	100.00%	99.53%	93.33%
JEV Noul validation-calibrated	JEV	99.90%	99.63%	100.00%	99.81%	97.33%
GPT-5.6 Sol	LLM	99.93%	99.72%	100.00%	99.86%	98.00%

Estimated marginal inference cost for 1,000 transcripts

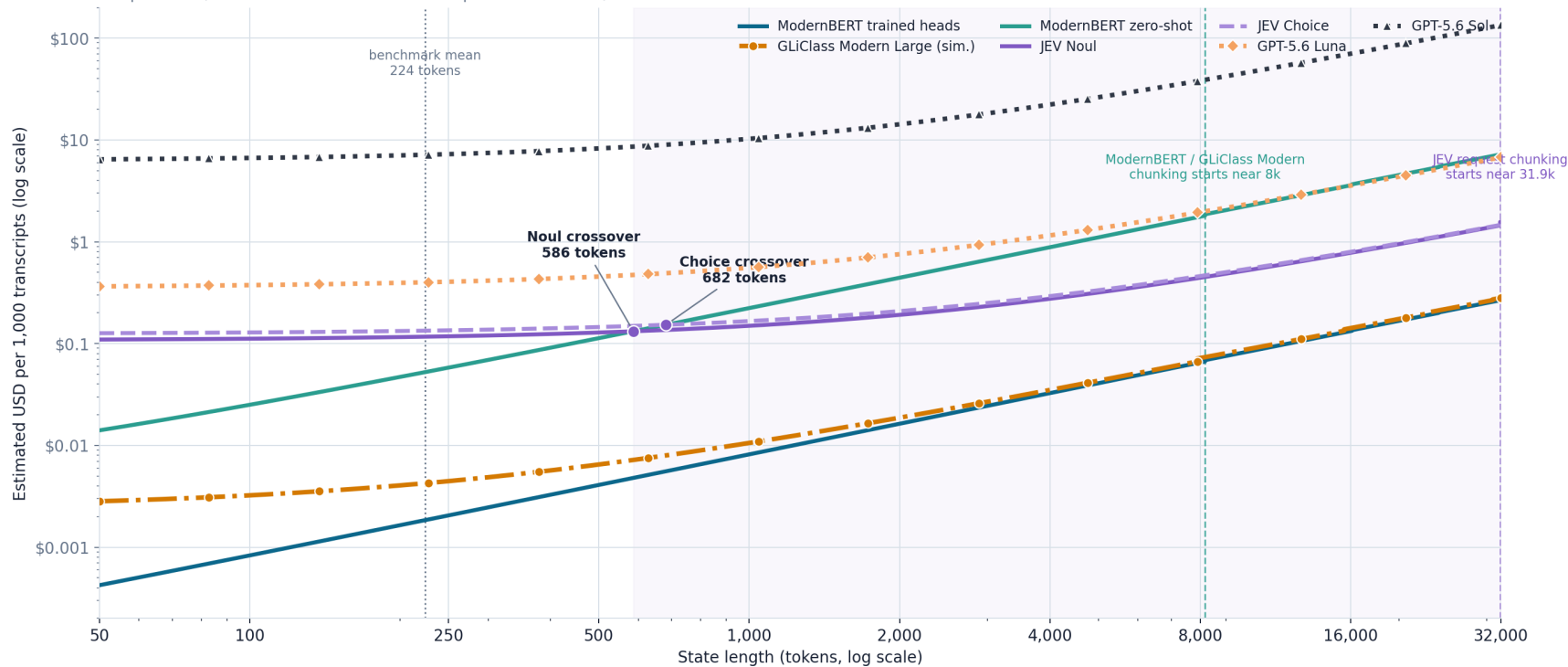
USD API charges · logarithmic scale above \$0.03 · validation/training and local hardware excluded



* Local encoders have no API fee; electricity, hardware, hosting, and operations are not zero and are not estimated. Sol/Luna use a standardized compact prompt estimate and exclude hidden reasoning tokens.

Estimated cost per 1,000 transcripts by state length

27 questions / labels · 256-token chunk overlap · H100 at USD 5/hour · lower cost is better

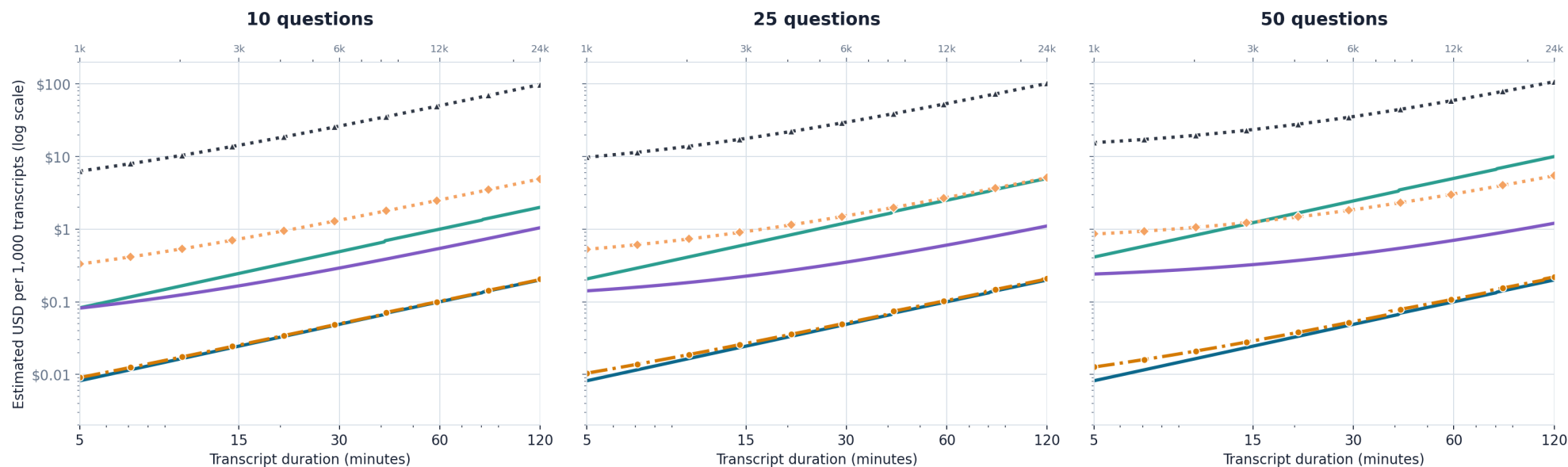


Each curve is the estimated cost of 1,000 transcripts. Sol/Luna include standardized input plus 212 output tokens; hidden reasoning tokens are excluded. GLiClass Modern uses an H100 throughput proxy, not a measured H100 result. Long-context throughput changes are not modeled.

Transcript classification cost by duration and taxonomy size

Top axes show state tokens · 200 state tokens/minute scenario · fixed encoder heads versus runtime questions

Fixed trained encoder Pairwise zero-shot encoder Single-call LLM: Luna
 Shared-state zero-shot encoder Hosted shared-state billing Single-call LLM: Sol

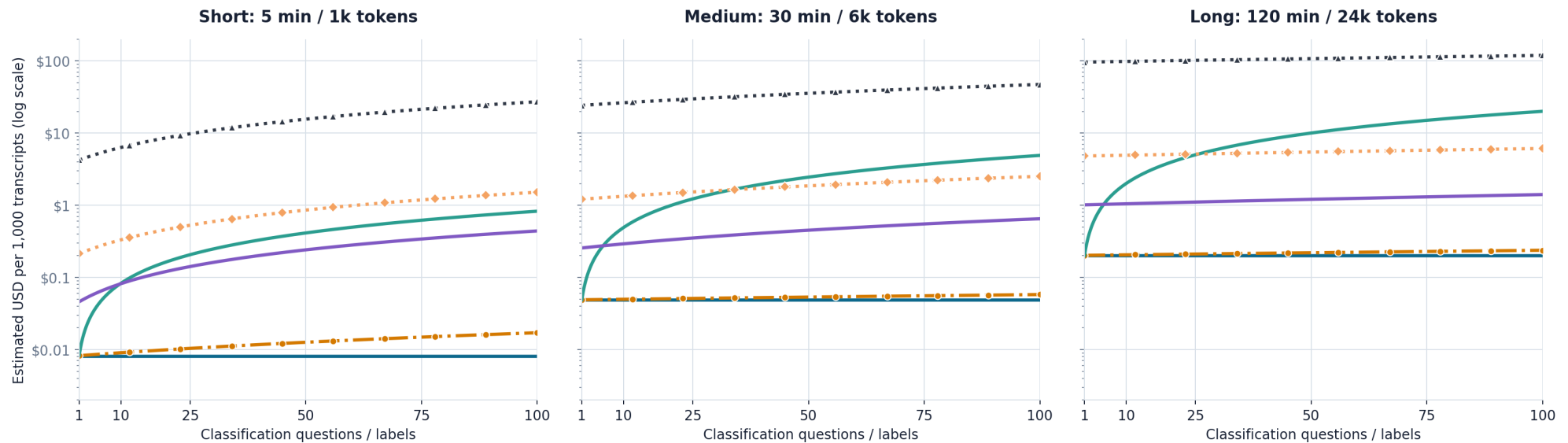


Question overhead is scaled proportionally from the measured 27-question workload. Fixed trained heads require prior training; JEV shows billed tokens, not verified internal compute. Hidden LLM reasoning tokens are excluded.

Transcript classification cost as the taxonomy grows

Short, medium, and long transcript scenarios · pairwise NLI repeats the state for every runtime question

Fixed trained encoder Pairwise zero-shot encoder Single-call LLM: Luna
Shared-state zero-shot encoder Hosted shared-state billing Single-call LLM: Sol



Fixed trained-head cost is nearly flat because head compute is treated as negligible. Runtime-label systems scale label, billing, or output overhead without repeating the full state in the external cost model.

Cost formulas and assumptions

Estimated USD per 1,000 transcripts

Definitions

S = state tokens q = 27 questions / labels overlap = 256 tokens
kM = max(1, ceil((S - 256)/(8178.11 - 256))) kGM = max(1, ceil((S - 256)/(7895 - 256)))
kGL = max(1, ceil((S - 256)/(357 - 256))) kJ = max(1, ceil((S - 256)/(31890 - 256)))

ModernBERT trained heads	$0.008154 * [S + 256*(kM - 1) + 2*kM] / 1,000$
GLiClass Modern Large	$0.008154 * [S + 256*(kGM - 1) + 297*kGM] / 1,000$
GLiClass Large (2 groups)	$0.012669 * \{2*[S + 256*(kGL - 1)] + 294*kGL\} / 1,000$
ModernBERT zero-shot	$0.008154 * \{27*[S + 256*(kM - 1)] + 375*kM\} / 1,000$
JEV Noul	$0.042 * [S + 256*(kJ - 1) + 2559.74*kJ] / 1,000$
JEV Choice	$0.042 * [S + 256*(kJ - 1) + 2963.67*kJ] / 1,000$
GPT-5.6 Luna	$[0.20*(S + 503.53) + 1.20*212] / 1,000$
GPT-5.6 Sol	$[4.00*(S + 503.53) + 20.00*212] / 1,000$

Rates. ModernBERT and GLiClass Modern use the 170.3k processed-token/s H100 proxy. JEV uses USD 0.042/M input. Luna uses USD 0.20/M input and USD 1.20/M output; Sol uses USD 4/M input and USD 20/M output.

Assumptions. GLiClass Large uses a 109.6k token/s proxy. Sol/Luna use S + 503.53 input tokens and 212 output tokens; hidden reasoning tokens are excluded. Neither GLiClass checkpoint was measured on H100 here.

What is shared - and what is fixed

The low-cost trained encoder and arbitrary-question JEV solve different serving problems

ModernBERT paths

Zero-shot NLI: N state/question pairs. Measured token amplification: 28.67x for 27 questions.

- It accepts arbitrary hypotheses, but repeats the state inside each premise/hypothesis sequence.
- Trained heads encode the state once, then apply 27 logistic functions.
- The trained path has no question tokens at inference; label meaning is stored in fixed head weights.
- Above about 8.2k state tokens, both paths require overlapping chunks and probability aggregation.
- A novel question requires a new or retrained head.

JEV evidence

- The API accepts arbitrary state and arbitrary typed questions together.
- A state can approach 32k tokens minus its longest question; larger states require multiple requests.
- Billing behaves like state + N x question + fixed overhead.
- Latency was approximately flat from 1 to 16 questions.
- TypeSafe claims a new architecture, parallel sampler, and RLCD training.
- Public materials do not disclose the compute graph or prove state is encoded once.

Conclusion: JEV's arbitrary-question, shared-state-like product behavior is real and useful. It is not enough to verify architectural novelty. Ordinary GPU batching can hide repeated state compute at small N, and known dual-encoder, cached-state, late-interaction, or shared-memory designs could expose similar behavior.

Methodology and controls

What was held fixed and what changed

Dataset

- 1,000 synthetic alternating Agent/Caller conversations.
- 27 binary attributes per transcript.
- Scenario-family-stratified split: 700 / 150 / 150.
- Test truth excluded from training, prompt refinement, and threshold selection.

Metrics

- Micro accuracy, precision, recall, and F1 over 4,050 decisions.
- Exact match requires all 27 labels in a transcript to be correct.

Generic cost projections

- Duration scenarios assume 200 state tokens per minute.
- Duration panels use 10, 25, and 50 questions; taxonomy panels span 1 to 100.
- Executive winner maps use 1k to 32k state tokens, 10 to 100 questions, and 10% to 100% paid H100 utilization.
- Question overhead scales proportionally from the measured 27-question workload.

Encoder variants

- Zero-shot: fixed premise/hypothesis NLI probabilities.
- GLiClass: uni-encoder label scoring; Modern v3 used one pass, while the 512-token Large v3 used two label groups to prevent truncation.
- Optimized threshold: 27 validation-selected scalar cutoffs; no weight training.
- Trained: frozen 1,024-d embedding plus 27 logistic heads and validation thresholds.

JEV and LLM variants

- JEV sends 27 typed questions in one request per transcript.
- Final Noul uses structured speakers and strict true/false boundaries.
- Sol and Luna produced blind binary JSON predictions before truth was joined.

Key distinction: trained ModernBERT learned approximately 27,675 logistic-head parameters from 700 rows. The optimized-threshold variant learned only 27 cutoffs from validation. The difference between their test results therefore estimates the value of supervised decision boundaries beyond operating-point calibration.

Conclusions, limitations, and sources

Practical interpretation

- For the highest measured quality, Sol led by one decision over calibrated JEV Noul, but its standardized benchmark-mean API estimate is \$7.152 per 1,000 transcripts versus \$0.400 for Luna and \$0.117 for JEV Noul.
- GLiClass Modern combines arbitrary runtime labels with a simulated \$0.0043 cost per 1,000 benchmark-mean transcripts, versus \$0.0525 for zero-shot ModernBERT and \$0.1169 for JEV Noul.
- Trained ModernBERT was by far the cheapest path, but only because its 27 questions were converted into fixed supervised heads.
- GLiClass Modern Large improved the calibrated zero-shot open-encoder result to 97.90% accuracy, but remained below trained ModernBERT.
- Threshold calibration is worthwhile, but it does not replace supervised heads when enough labeled examples exist.

Limitations

- Synthetic templates are more explicit and regular than production conversations.
- Only 150 transcripts were held out; attribute decisions within a transcript are correlated.
- Choice versus Noul changed multiple factors and is not a clean primitive-only comparison.
- The H100 scenario uses throughput proxies and holds token throughput constant across state lengths; GLiClass Modern inherits the ModernBERT-large proxy and GLiClass Large scales it using an official A6000 relative-throughput ratio.
- JEV cost is extrapolated from billing; hosted capacity at the equivalent throughput was not tested.
- Sol/Luna standardized cost estimates exclude unavailable hidden reasoning-token usage.
- Sol/Luna state scaling assumes one added LLM input token per ModernBERT state-axis token; tokenizer differences make this approximate.
- GLiClass Large used two label groups while GLiClass Modern Large used one; their local timings are not a one-pass checkpoint comparison.

Sources

- TypeSafe API: <https://docs.typesafe.ai/api>
- TypeSafe model limits: <https://docs.typesafe.ai/models>
- TypeSafe JEV launch/pricing: <https://typesafe.ai/blog/introducing-system-one-models-and-jev>
- ModernBERT documentation: https://huggingface.co/docs/transformers/en/model_doc/modernbert
- H100 ModernBERT-base observation: https://www.linkedin.com/posts/michael-feil_the-latest-release-of-infinity-https%3A%2F%2Fwww.linkedin.com%2Factivity-7280971190632943616-E07N
- OpenAI Sol pricing: <https://developers.openai.com/api/docs/models/gpt-5.6-sol>
- OpenAI Luna pricing: <https://developers.openai.com/api/docs/models>
- GLiClass Modern Large v3: <https://huggingface.co/knowledgator/gliclass-modern-large-v3.0>
- GLiClass Large v3: <https://huggingface.co/knowledgator/gliclass-large-v3.0>